Llamada API Copernicus Marine (Variables: Temperatura, corriente superficiales Este y Norte):

```
import copernicusmarine

copernicusmarine.subset(
   dataset_id="cmems_mod_glo_phy_my_0.083deg_P1D-m",
   variables=["thetao", "uo", "vo"],
   minimum_longitude=-19,
   maximum_longitude=-13,
   minimum_latitude=27,
   maximum_latitude=30,
   start_datetime="2023-08-21T00:00:00",
   end_datetime="2023-08-25T00:00:00",
   minimum_depth=0.49402499198913574,
   maximum_depth=0.49402499198913574,
)
```

Llamada API Copernicus Marine (Variables: Corriente de viento, año 2017):

```
import cdsapi

dataset = "reanalysis-era5-single-levels"
request = {
   "product_type": ["reanalysis"],
   "variable": [
      "10m_u_component_of_wind",
      "10m_v_component_of_wind"
   ],
   "year": ["2017"],
   "month": [
      "06", "07", "08",
      "09", "10"
   ],
   "day": [
      "01", "02", "03",
      "04", "05", "06",
      "07", "08", "09",
      "10", "11", "12",
      "13", "14", "15",
      "16", "17", "18",
      "19", "20", "21",
      "22", "23", "24",
      "25", "26", "27",
      "28", "29", "30",
      "31"
```

```
    ],
    "time": [
        "00:00", "01:00", "02:00",
        "03:00", "04:00", "05:00",
        "06:00", "07:00", "08:00",
        "09:00", "10:00", "11:00",
        "12:00", "13:00", "14:00",
        "15:00", "16:00", "17:00",
        "18:00", "19:00", "20:00",
        "21:00", "22:00", "23:00"
    ],
    "data_format": "netcdf",
    "download_format": "zip",
    "area": [29.5, -18.5, 27.25, -13.25]
}

client = cdsapi.Client()
client.retrieve(dataset, request).download()
```

Llamada API Copernicus Marine (Variables: Corriente de viento, año 2022):

```
import cdsapi

dataset = "reanalysis-era5-single-levels"
request = {
    "product_type": ["reanalysis"],
    "variable": [
        "10m_u_component_of_wind",
        "10m_v_component_of_wind"
    ],
    "year": ["2022"],
    "month": ["09"],
    "day": [
        "01", "02", "03",
        "04", "05", "06",
        "07", "08", "09",
        "10", "11", "12",
        "13", "14", "15",
        "16", "17", "18",
        "19", "20", "21",
        "22", "23", "24",
        "25", "26", "27",
        "28", "29", "30"
    ],
    "time": [
        "00:00", "01:00", "02:00",
        "03:00", "04:00", "05:00",
```

```
        "06:00", "07:00", "08:00",
        "09:00", "10:00", "11:00",
        "12:00", "13:00", "14:00",
        "15:00", "16:00", "17:00",
        "18:00", "19:00", "20:00",
        "21:00", "22:00", "23:00"
    ],
    "data_format": "grib",
    "download_format": "unarchived",
    "area": [29.5, -18.5, 27.25, -13.25]
}

client = cdsapi.Client()
client.retrieve(dataset, request).download()
```

Llamada API Copernicus Marine (Variables: Corriente de viento, año 2023):

```
import cdsapi

dataset = "reanalysis-era5-single-levels"
request = {
    "product_type": ["reanalysis"],
    "variable": [
        "10m_u_component_of_wind",
        "10m_v_component_of_wind"
    ],
    "year": ["2023"],
    "month": ["08", "09", "10"],
    "day": [
        "01", "02", "03",
        "04", "05", "06",
        "07", "08", "09",
        "10", "11", "12",
        "13", "14", "15",
        "16", "17", "18",
        "19", "20", "21",
        "22", "23", "24",
        "25", "26", "27",
        "28", "29", "30",
        "31"
    ],
    "time": [
        "00:00", "01:00", "02:00",
        "03:00", "04:00", "05:00",
        "06:00", "07:00", "08:00",
        "09:00", "10:00", "11:00",
        "12:00", "13:00", "14:00",
```

```
        "15:00", "16:00", "17:00",
        "18:00", "19:00", "20:00",
        "21:00", "22:00", "23:00"
    ],
    "data_format": "grib",
    "download_format": "unarchived",
    "area": [29.5, -18.5, 27.25, -13.25]
}

client = cdsapi.Client()
client.retrieve(dataset, request).download()
```

Llamada API Copernicus Marine (Variables: Hierro disuelto, año 2017):

```
import copernicusmarine

copernicusmarine.subset(
  dataset_id="cmems_mod_glo_bgc_my_0.25deg_P1M-m",
  variables=["fe"],
  minimum_longitude=-13.25,
  maximum_longitude=-18.5,
  minimum_latitude=27.25,
  maximum_latitude=29.5,
  start_datetime="2017-06-01T00:00:00",
  end_datetime="2017-10-31T00:00:00",
  minimum_depth=0.5057600140571594,
  maximum_depth=0.5057600140571594,
)
```

Llamada API Copernicus Marine (Variables: Hierro disuelto, año 2022):

```
import copernicusmarine

copernicusmarine.subset(
  dataset_id="cmems_mod_glo_bgc_my_0.25deg_P1M-m",
  variables=["fe"],
  minimum_longitude=-13.25,
  maximum_longitude=-18.5,
  minimum_latitude=27.25,
  maximum_latitude=29.5,
  start_datetime="2022-09-01T00:00:00",
  end_datetime="2022-09-30T00:00:00",
  minimum_depth=0.5057600140571594,
  maximum_depth=0.5057600140571594,
)
```

Llamada API Copernicus Marine (Variables: Hierro disuelto, año 2023):

```
import copernicusmarine

copernicusmarine.subset(
  dataset_id="cmems_mod_glo_bgc_my_0.25deg_P1M-m",
  variables=["fe"],
  minimum_longitude=-13.25,
  maximum_longitude=-18.5,
  minimum_latitude=27.25,
  maximum_latitude=29.5,
  start_datetime="2023-08-01T00:00:00",
  end_datetime="2023-10-31T00:00:00",
  minimum_depth=0.5057600140571594,
  maximum_depth=0.5057600140571594,
)
```

Llamada API Copernicus Marine (Variables: velocidad vertical del agua, año 2017):

```
import copernicusmarine

copernicusmarine.subset(
  dataset_id="cmems_mod_glo_phy-wcur_anfc_0.083deg_P1M-m",
  variables=["wo"],
  minimum_longitude=-13.25,
  maximum_longitude=-18.5,
  minimum_latitude=27.25,
  maximum_latitude=29.5,
  start_datetime=""2017-06-01T00:00:00",
  end_datetime= 2017-10-31T00:00:00",
  minimum_depth=0.49402499198913574,
  maximum_depth=0.49402499198913574,
)
```

Llamada API Copernicus Marine (Variables: velocidad vertical del agua, año 2022):

```
import copernicusmarine

copernicusmarine.subset(
  dataset_id="cmems_mod_glo_phy-wcur_anfc_0.083deg_P1M-m",
  variables=["wo"],
  minimum_longitude=-13.25,
  maximum_longitude=-18.5,
  minimum_latitude=27.25,
  maximum_latitude=29.5,
```

```
  start_datetime="2022-09-01T00:00:00",
  end_datetime="2022-09-30T00:00:00",
  minimum_depth=0.49402499198913574,
  maximum_depth=0.49402499198913574,
)
```

Llamada API Copernicus Marine (Variables: velocidad vertical del agua, año 2023):

```
import copernicusmarine

copernicusmarine.subset(
  dataset_id="cmems_mod_glo_phy-wcur_anfc_0.083deg_P1M-m",
  variables=["wo"],
  minimum_longitude=-13.25,
  maximum_longitude= -18.5,
  minimum_latitude=27.25,
  maximum_latitude=29.5,
  start_datetime="2023-08-01T00:00:00",
  end_datetime="2023-10-31T00:00:00",
  minimum_depth=0.49402499198913574,
  maximum_depth=0.49402499198913574,
)
```

Llamada API Copernicus Marine (Variables Deposición de polvo mineral sahariano :año 2017):

```
import cdsapi

dataset = "cams-global-reanalysis-eac4"
request = {
    "variable": [
        "dust_aerosol_optical_depth_550nm",
        "organic_matter_aerosol_optical_depth_550nm",
        "particulate_matter_1um",
        "particulate_matter_2.5um",
        "particulate_matter_10um",
        "sulphate_aerosol_optical_depth_550nm",
        "total_aerosol_optical_depth_469nm",
        "total_aerosol_optical_depth_550nm",
        "total_aerosol_optical_depth_670nm",
        "total_aerosol_optical_depth_865nm",
        "total_aerosol_optical_depth_1240nm"
    ],
    "date": ["2017-06-01/2017-10-31"],
    "time": [
        "00:00", "03:00", "06:00",
```

```
        "09:00", "12:00", "15:00",
        "18:00", "21:00"
    ],
    "data_format": "grib",
    "area": [29.5, -18.5, 27.25, -13.25]
}

client = cdsapi.Client()
client.retrieve(dataset, request).download()
```

Llamada API Copernicus Marine (Variables Deposición de polvo mineral sahariano :año 2022):

```
import cdsapi

dataset = "cams-global-reanalysis-eac4"
request = {
    "variable": [
        "dust_aerosol_optical_depth_550nm",
        "organic_matter_aerosol_optical_depth_550nm",
        "particulate_matter_1um",
        "particulate_matter_2.5um",
        "particulate_matter_10um",
        "sulphate_aerosol_optical_depth_550nm",
        "total_aerosol_optical_depth_469nm",
        "total_aerosol_optical_depth_550nm",
        "total_aerosol_optical_depth_670nm",
        "total_aerosol_optical_depth_865nm",
        "total_aerosol_optical_depth_1240nm"
    ],
    "date": ["2022-09-01/2022-09-30"],
    "time": [
        "00:00", "03:00", "06:00",
        "09:00", "12:00", "15:00",
        "18:00", "21:00"
    ],
    "data_format": "grib",
    "area": [29.5, -18.5, 27.25, -13.25]
}

client = cdsapi.Client()
client.retrieve(dataset, request).download()
```

Llamada API Copernicus Marine (Variables Deposición de polvo mineral sahariano :año 2023):

```
import cdsapi

dataset = "cams-global-reanalysis-eac4"
request = {
    "variable": [
        "dust_aerosol_optical_depth_550nm",
        "organic_matter_aerosol_optical_depth_550nm",
        "particulate_matter_1um",
        "particulate_matter_2.5um",
        "particulate_matter_10um",
        "sulphate_aerosol_optical_depth_550nm",
        "total_aerosol_optical_depth_469nm",
        "total_aerosol_optical_depth_550nm",
        "total_aerosol_optical_depth_670nm",
        "total_aerosol_optical_depth_865nm",
        "total_aerosol_optical_depth_1240nm"
    ],
    "date": ["2023-08-01/2023-10-31"],
    "time": [
        "00:00", "03:00", "06:00",
        "09:00", "12:00", "15:00",
        "18:00", "21:00"
    ],
    "data_format": "grib",
    "area": [29.5, -18.5, 27.25, -13.25]
}

client = cdsapi.Client()
client.retrieve(dataset, request).download()
```